Example 5.6 Show that the points $A(7,10)$, $B(-2,5)$, $C(3,-4)$ are the vertices of a right angled triangle.

Solution

Here $A = (7, 10)$, $B = (-2, 5)$, $C = (3,-4)$

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(-2 - 7)^2 + (5 - 10)^2}$$

$$= \sqrt{(-9)^2 + (-5)^2}$$

$$= \sqrt{(81 + 25)}$$

$$= \sqrt{106}$$

$$AB^2 = 106 \quad \ldots (1)$$

$$BC = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(3 - (-2))^2 + (-4 - 5)^2} = \sqrt{(5)^2 + (-9)^2}$$

$$= \sqrt{25 + 81} = \sqrt{106}$$

$$BC^2 = 106 \quad \ldots (2)$$

$$AC = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(3 - 7)^2 + (-4 - 10)^2} = \sqrt{(-4)^2 + (-14)^2}$$

$$= \sqrt{16 + 196} = \sqrt{212}$$

$$AC^2 = 212 \quad \ldots (3)$$

From (1), (2) & (3) we get,

$$AB^2 + BC^2 = 106 + 106 = 212 = AC^2$$

Since $AB^2 + BC^2 = AC^2$

$\therefore$ ΔABC is a right angled triangle, right angled at B.

Right angled triangle

We know that the sum of the squares of two sides is equal to the square of the third side,which is the hypotenuse of a right angled triangle.

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